

AP CALCULUS AB – UNIT 2

2.3 The Derivative: Graphs, Differentiability, and Continuity



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Estimate a derivative from a graph, numerical table, or local data.
- Compare forward, backward, and centered differences.
- Judge the reliability of an estimate using interval size and local smoothness.
- Explain why differentiability at a point implies continuity there.
- Identify discontinuities, corners, cusps, and vertical tangents as sources of nondifferentiability.
- Determine parameter values that make a piecewise function continuous and differentiable.

Keys: Essential Knowledge and Vocabulary

Derivative Estimation: $\frac{f(a+h) - f(a)}{h}$.
Forward difference

Derivative Estimation: $\frac{f(a) - f(a-h)}{h}$.
Backward difference

Derivative Estimation: $\frac{f(a+h) - f(a-h)}{2h}$, often more accurate for smooth functions.
Centered difference

Derivative Estimation: Approximate the slope of a carefully chosen tangent line, then use rise over run.
Graphical estimate

Differentiability and Continuity: Necessary condition If $f'(a)$ exists, then f is continuous at a .

Differentiability and Continuity: Not sufficient Continuity alone does not guarantee differentiability; $|x|$ is continuous but has a corner at 0.

Differentiability and Continuity: One-sided derivatives $f'(a)$ exists only when the left and right derivative limits exist finitely and are equal.

Differentiability and Continuity: Common failures Discontinuity, corner, cusp, vertical tangent, or sufficiently wild oscillation.

Explanation, Strategy, and Common Errors

- **Derivative Estimation.** Use points close to the target, but not so close that rounding error dominates. When a table is rounded, several nearby estimates should be compared.
- **Derivative Estimation.** A centered difference balances information from both sides and cancels some

first-order error for smooth functions.

- **Derivative Estimation.** At a corner or cusp, left and right slope estimates approach different values or become unbounded; reporting one centered number can hide nondifferentiability.
- **Differentiability and Continuity.** To test a piecewise join, first impose continuity. Only after the function values match should you impose equality of the one-sided derivatives.
- **Differentiability and Continuity.** A vertical tangent may have a visually well-defined tangent line but its slope is not a finite real number, so the ordinary derivative does not exist.
- **Differentiability and Continuity.** When justifying, name the exact condition that fails: unequal one-sided slopes, unbounded slope, or lack of continuity.

Solved Example: Derivative Estimation - Centered estimate

Given $f(1.9) = 4.61$ and $f(2.1) = 5.43$, estimate $f'(2)$.

Answer and Reasoning

$$\frac{5.43 - 4.61}{0.2} = 4.1.$$

Solved Example: Derivative Estimation - From a graph

A tangent line through the point of interest also passes approximately through $(1, 2.2)$ and $(5, 7.0)$. Estimate the derivative.

Answer and Reasoning

$$\frac{7.0 - 2.2}{5 - 1} = 1.2.$$

Solved Example: Derivative Estimation - Detecting a corner

Near $x = 0$, left secant slopes approach -2 while right secant slopes approach 3 . What is $f'(0)$?

Answer and Reasoning

It does not exist because the one-sided derivative limits disagree.

Solved Example: Differentiability and Continuity - Continuous but not differentiable

Analyze $f(x) = |x - 2|$ at $x = 2$.

Answer and Reasoning

The function is continuous. The left derivative is -1 and the right derivative is 1 , so $f'(2)$ does not exist.

Solved Example: Differentiability and Continuity - Piecewise differentiability

Find m, b so $f(x) = mx + b$ for $x < 1$ and $f(x) = x^2 + 1$ for $x \geq 1$ is differentiable at 1 .

Answer and Reasoning

Continuity gives $m + b = 2$; slope matching gives $m = 2$. Hence $b = 0$.

Solved Example: Differentiability and Continuity - Vertical tangent

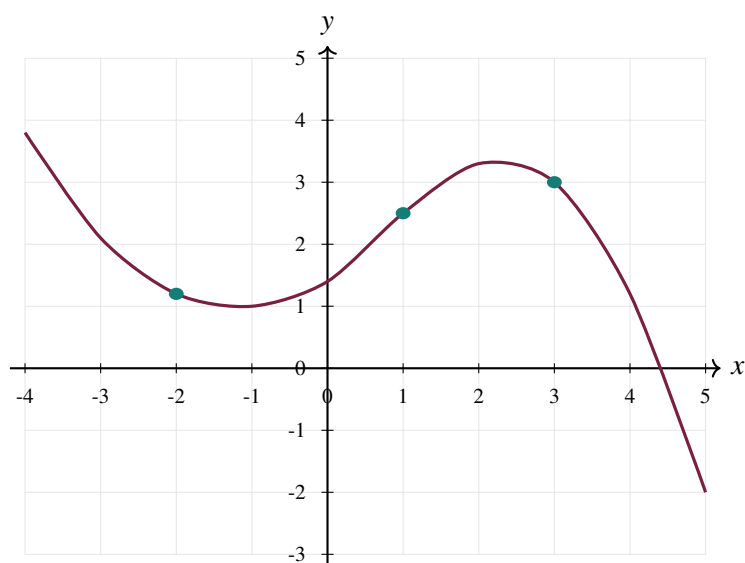
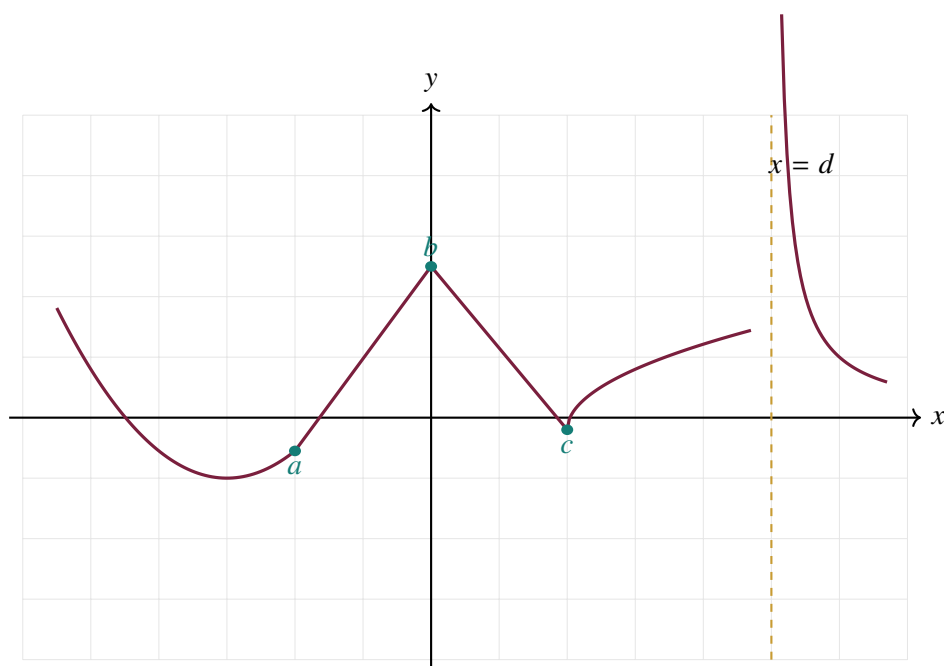
For $f(x) = \sqrt[3]{x}$, discuss differentiability at 0.

Answer and Reasoning

The difference quotient is $h^{-2/3} \rightarrow +\infty$, so there is a vertical tangent and no finite derivative at 0.

Shared Representation / Data**Derivative Estimation**

x	1.80	1.90	1.99	2.00	2.01	2.10	2.20
$f(x)$	4.812	4.947	5.085	5.100	5.115	5.247	5.388

**Differentiability and Continuity**

AP Exam Language

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. Derivative Estimation: Use $f(2.9) = 8.41$ and $f(3.1) = 9.61$ to estimate $f'(3)$ by a centered difference.

2. Derivative Estimation: Use $g(4) = 11.2$ and $g(4.05) = 11.38$ to make a forward estimate of $g'(4)$.

3. Derivative Estimation: Use $p(7) = 5.0$ and $p(6.9) = 5.27$ to make a backward estimate of $p'(7)$.

4. Derivative Estimation: A tangent line rises 5.4 units over a horizontal run of 3. Estimate the derivative.

5. Derivative Estimation: What units does an estimated derivative of revenue (dollars) with respect to units sold have?

6. Differentiability and Continuity: True or false: If f is differentiable at a , then it is continuous at a .

7. Differentiability and Continuity: True or false: If f is continuous at a , then it is differentiable at a .

8. Differentiability and Continuity: Name the failure at a jump discontinuity.

9. Differentiability and Continuity: For $f(x) = |x + 3|$, where is f not differentiable?

10. Differentiability and Continuity: For $f(x) = x^{2/3}$, what feature occurs at 0?

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

11. Derivative Estimation: Using the shared table, estimate $f'(2)$ with $h = 0.01$ and with $h = 0.10$.

12. Derivative Estimation: From the shared graph, rank the derivative estimates at $x = -2, 1, 3$ from least to greatest.

13. Derivative Estimation: At $x = 5$, forward and backward estimates are 2.14 and 2.06. Give a centered estimate.

14. Derivative Estimation: A table has values rounded to the nearest tenth. Explain why using $h = 0.0001$ may be unreliable.

15. Derivative Estimation: Values near a produce centered estimates $-1.8, -1.95, -1.995$. Estimate $f'(a)$.

16. Differentiability and Continuity: Determine whether $f(x) = \begin{cases} x + 1, & x < 0 \\ x^2 + 1, & x \geq 0 \end{cases}$ is differentiable at 0.

17. Differentiability and Continuity: Find k so $f(x) = \begin{cases} kx + 2, & x < 1 \\ x^2 + k, & x \geq 1 \end{cases}$ is continuous at 1.

18. Differentiability and Continuity: For $f(x) = \sqrt{|x|}$, classify the behavior at 0.

19. Differentiability and Continuity: A graph has a removable hole at $x = 4$ but is redefined at the limiting value. Is differentiability automatic?

20. Differentiability and Continuity: Explain why a tangent slope cannot exist at an isolated point of a graph.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

21. Derivative Estimation: For a smooth function, forward estimates with $h = .2, .1, .05$ are 3.40, 3.20, 3.10. Predict the derivative and explain the pattern.

22. Derivative Estimation: Left slope estimates approach $-\infty$ and right slope estimates approach $+\infty$. What geometric feature is suggested?

23. Derivative Estimation: Given $f(a - h) = 7.92$, $f(a) = 8.00$, $f(a + h) = 8.10$, compare forward, backward, and centered estimates in terms of h .

24. Derivative Estimation: The graph of f near $x = 2$ looks linear with slope -4 , but has an open circle at $x = 2$ and a filled point elsewhere. Can $f'(2)$ exist?

25. Differentiability and Continuity: Find a, b so $f(x) = \begin{cases} ax + b, & x \leq 2 \\ x^2 - 1, & x > 2 \end{cases}$ is differentiable at 2.

26. Differentiability and Continuity: Determine differentiability at 0 for $f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$.

27. Differentiability and Continuity: Determine differentiability at 0 for $g(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$.

28. Differentiability and Continuity: A function has a vertical tangent at a and is continuous there. Explain why continuity does not settle differentiability.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

29. Derivative Estimation: Use $f(a - 2h) = 4.7$, $f(a - h) = 4.9$, $f(a + h) = 5.5$, $f(a + 2h) = 6.0$ to compare centered estimates with radii h and $2h$.

30. Derivative Estimation: A symmetric estimate is exactly 0 for every available h . Does this prove $f'(a) = 0$? Explain.

31. Derivative Estimation: At a , centered differences converge to 2, while forward differences converge to 5. What conclusion is forced?

32. Differentiability and Continuity: Find p, q so $f(x) = \begin{cases} px^2 + q, & x < 1 \\ 3x - p, & x \geq 1 \end{cases}$ is differentiable at 1.

33. Differentiability and Continuity: For what $\alpha > 0$ is $f(x) = |x|^\alpha$ differentiable at 0?

34. Differentiability and Continuity: Can a function be differentiable on (a, b) but discontinuous at a ? Explain.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

35. Derivative Estimation: Show algebraically that the centered difference for $f(x) = x^3$ at a equals $3a^2 + h^2$.

36. Derivative Estimation: A function satisfies $f(a + h) = f(a) + mh + h^2 \sin(1/h)$ for $h \neq 0$. Determine what forward numerical estimates approach.

37. Differentiability and Continuity: Construct a function continuous at 0 whose derivative fails because of oscillation, not a corner or vertical tangent.

38. Differentiability and Continuity: Prove differentiability implies continuity using $f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a)$.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

- 1. Derivative Estimation:** Which estimate generally uses data from both sides of a ?
- (A) forward difference (B) backward difference
(C) centered difference (D) average function value
- 2. Derivative Estimation:** If left derivative estimates approach 4 and right derivative estimates approach 4, the best conclusion is
- (A) $f(a) = 4$ (B) $f'(a)$ is likely 4 if the function is defined appropriately
(C) the function has a corner (D) the derivative is 8
- 3. Derivative Estimation:** Tangent-line points $(2, -1)$ and $(6, 5)$ give derivative estimate
- (A) $\frac{2}{3}$ (B) $\frac{3}{2}$
(C) 4 (D) 6
- 4. Derivative Estimation:** Which situation most threatens a numerical derivative estimate?
- (A) a smooth graph and balanced data (B) a table rounded coarsely with extremely small h
(C) using correct units (D) using points near the target
- 5. Differentiability and Continuity:** Which condition is sufficient to conclude continuity at a ?
- (A) $f'(a)$ exists (B) $f(a)$ exists
(C) the graph has a tangent-looking line (D) left and right function values are finite
- 6. Differentiability and Continuity:** Which function is continuous but not differentiable at 0?
- (A) x^2 (B) x^3
(C) $|x|$ (D) e^x
- 7. Differentiability and Continuity:** At a cusp, one-sided derivative behavior is typically
- (A) equal finite values (B) opposite signed unbounded values
(C) zero on both sides (D) undefined because the function must be discontinuous
- 8. Differentiability and Continuity:** To make a two-piece function differentiable at a join, one must ensure
- (A) only equal function values (B) only equal one-sided slopes
(C) continuity and equal one-sided derivatives (D) the formulas are identical everywhere

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

9. Derivative Estimation: Using the shared table, the centered estimate of $f'(2)$ with $h = .1$ is

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|----------|---------|
| (A) 0.75 | (B) 1.5 |
| (C) 2.7 | (D) 5.1 |

10. Derivative Estimation: A calculator gives $f(4.001) = 9.004002$ and $f(3.999) = 8.996002$. The centered estimate at 4 is

- | | |
|-------|-------|
| (A) 2 | (B) 4 |
| (C) 8 | (D) 9 |

11. Differentiability and Continuity: A calculator graph shows a sharp corner at $x = 1.7$. The most appropriate conclusion is

- | | |
|--------------------------|---|
| (A) $f'(1.7) = 0$ | (B) $f'(1.7)$ does not exist if the corner is genuine |
| (C) f is discontinuous | (D) $f'(1.7) = 1.7$ |

12. Differentiability and Continuity: Numerical slopes near a are $-20, -100, -500$ from the left and $20, 100, 500$ from the right. This suggests

- | | |
|--------------------------|----------------------|
| (A) a horizontal tangent | (B) a jump |
| (C) a cusp | (D) a removable hole |

AP-Style Free-Response Practice

Calculator Permitted

Complete the following independent calculator-permitted problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Derivative Estimation. The table below gives selected values of a differentiable function g near $x = 2$.

x	1.8	1.9	2.0	2.1	2.2
$g(x)$	3.422	3.705	4.000	4.305	4.618

Find the forward and backward estimates of $g'(2)$ using $h = 0.1$.

[2 points]

(1b) Find the centered estimate using $h = 0.1$.

[2 points]

(1c) Explain why the centered estimate may be preferable here.

[2 points]

(2a) Problem 2: Differentiability and Continuity. Let $f(x) = \begin{cases} ax + b, & x < 2 \\ x^2 - 3x + 5, & x \geq 2. \end{cases}$

Find the condition on a, b for continuity at 2.

[2 points]

(2b) Find the condition on a for matching one-sided derivatives.

[2 points]

(2c) Find a, b that make f differentiable and justify.

[3 points]

No Calculator

Complete the following independent non-calculator problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Derivative Estimation. A graphing utility gives the following slopes of secants through $(1, f(1))$: 2.54, 2.504, 2.5004 for right increments 0.1, 0.01, 0.001, and 2.46, 2.496, 2.4996 for corresponding left increments.

Estimate $f'(1)$. [1 points]

(1b) Describe how the two sets of estimates support differentiability. [2 points]

(1c) Write a local tangent-line model if $f(1) = 7.2$. [2 points]

(2a) Problem 2: Differentiability and Continuity. A graphing utility is used to investigate $g(x) = |x - 1.35|^{2/3}$.

State the point at which differentiability should be investigated. [1 points]

(2b) Describe the expected one-sided slope behavior. [2 points]

(2c) Explain why a calculator display alone is not a proof, and give an analytic justification. [3 points]